



App Dev Club x Project Kuiper

Website

www.appdevclub.com

Dec 6th, 2024



Agenda

01

**Problem &
Solution**

02

Live Demo

03

**ML/Full Stack
Explanation**

04

**Team
Introductions**

Problem Overview

No Trust

- Lack of **communication**
- **Not following rules** set forward by FCC
- **Interference** with missions of Kuiper

Anomalous Satellite Observations

- Identifying anomalous satellites **behaving inconsistently** within their constellation.

Parking Satellite Probability

- Determining **key latent features** that give us the probability of a satellite performing an **orbit-raise maneuver**.

Solution

Monitoring

The whole goal of this platform is to make dashboard that can provide insights to your team in the most effective way.

Our solution is scalable and provides your team with the dashboards needed to provide insights.

Anomaly* Detection

Presents outlier satellites in a certain group based on the nine orbital elements.

Upon selection, graphs the selected orbital element compared to the normal behavior in that group.

Parking Prediction

Dashboard identifies satellites that are parking and shows:

- The probability of these satellites being in parking or in their operational orbit.
- How it has moved over a given span of time.



About Us

App Dev Club is a club at UMD that prepares students for industry software development. Our student teams develop softwares for Fortune 500 companies that are used every day.

Our team is comprised of 15 talented individuals, ranging from project leads, project managers, tech leads, engineers, and designers.



Our Approach: Machine Learning

ML Viability

The Idea

- Explore the viability of machine learning to identify satellite behaviours based on past satellite data.
- Identify the probability of a parked satellite performing a raise maneuver in the future.
- Identify anomalous satellites (behaving unusually) within launch groups.

What We Can Conclude

- Machine Learning can provide valuable insights into behaviours and trends.
- Models achieved high accuracy in distinguishing parked satellites from operational ones, offering insights into the transition likelihood of parked satellites.
- Clustering methods were very good at identifying satellites behaving unusually within launch groups, but were susceptible to false positives.
- Confidence metrics were introduced to enhance trust in model predictions.

ML Parking Model

The Strategy

In order to label historical data as parking or operational:

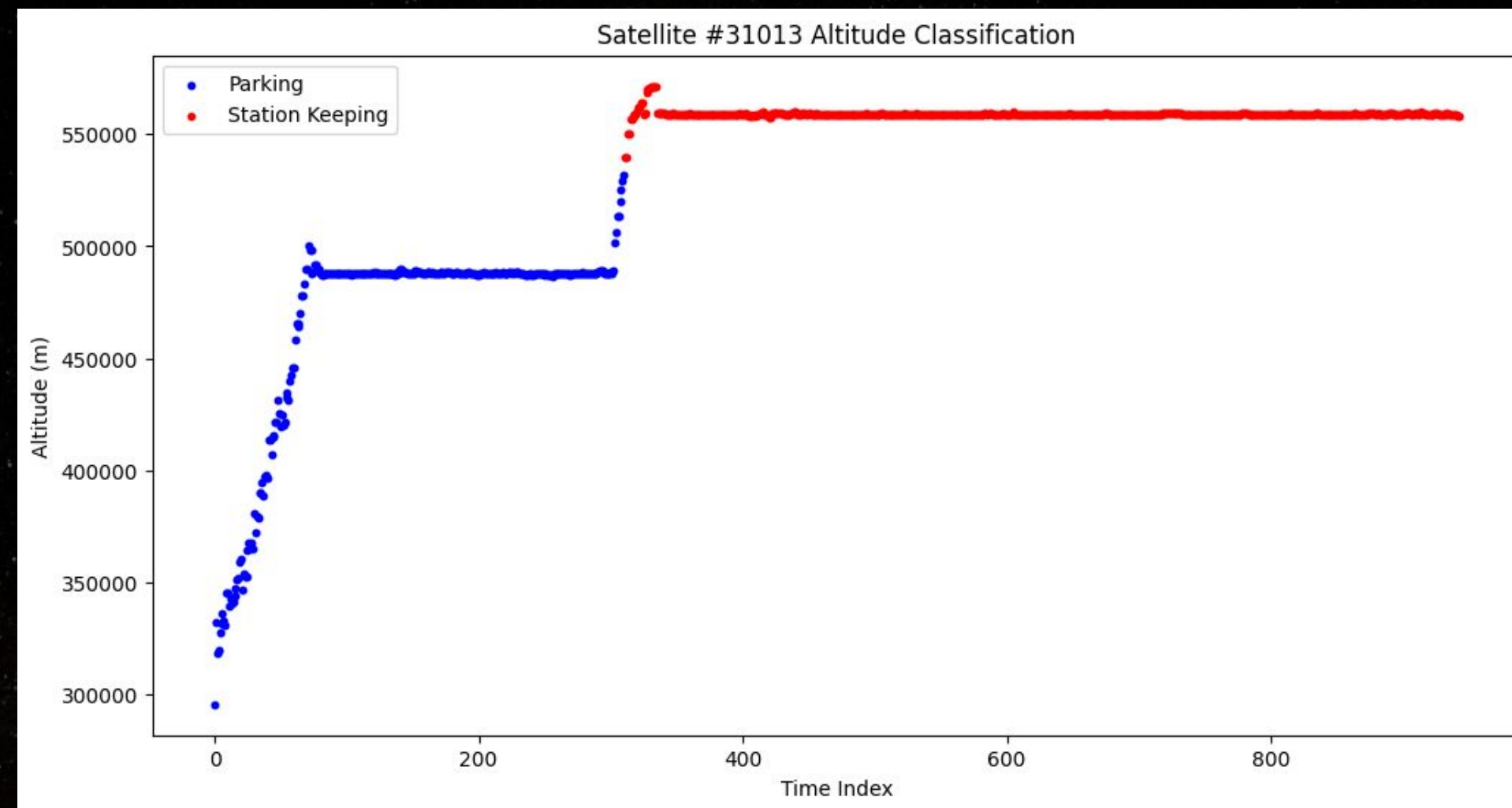
- Reframe the problem as: for some point in time, will there be a future point higher in altitude by a significant amount (x km)
- Compute x as the minimum value s.t $\geq 70\%$ of the satellite lifespan is labeled as operational
- After one point is labeled as operational, all in the future are

Our Data

- Open-source TLE data from SpaceTrack
- Easily accessible
- Has long-range historical data for variety of Satellites
- Focused on Starlink satellites



How it Performs

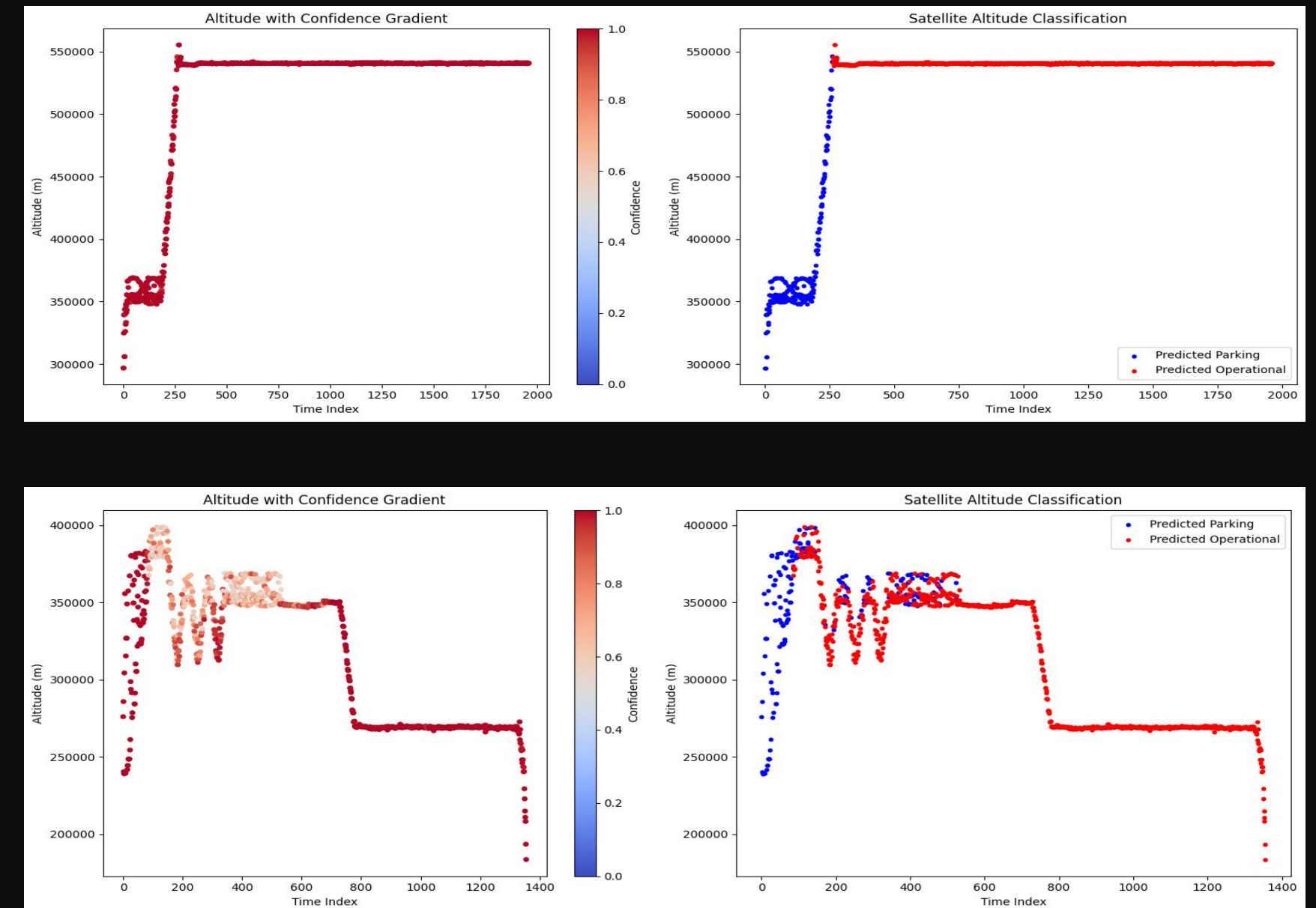




The Models

- Tried models that predict status based on a time index (XGBoost) as well as models that predict on a time window (C-LSTM, Transformer)
- Engineered features (e.g. the standard deviation of a satellite's altitude over the last few days) to help the model predict
- XGBoost learned patterns best, computationally efficient

Confidence Visualization: understand where the model is more uncertain in its prediction.



Accuracy

Overall Accuracy: 0.97	Percentage of Positive Predictions that were Correct	Percentage of Positive Points that were identified as Positive
Parking Status	0.90	0.89
Operational Status	0.98	0.98

Anomaly Detection Model

The Goal

Be able to detect satellites in a given group that are not behaving similarly to other satellites.

Allows for more informed decisions on satellite behaviors.

How?

Adapt classical outlier detection algorithms to time-series data.

Two-Fold approach: cross verification with both clustering and isolation forest methods.

The Data

Raw features include the orbital elements and velocity.

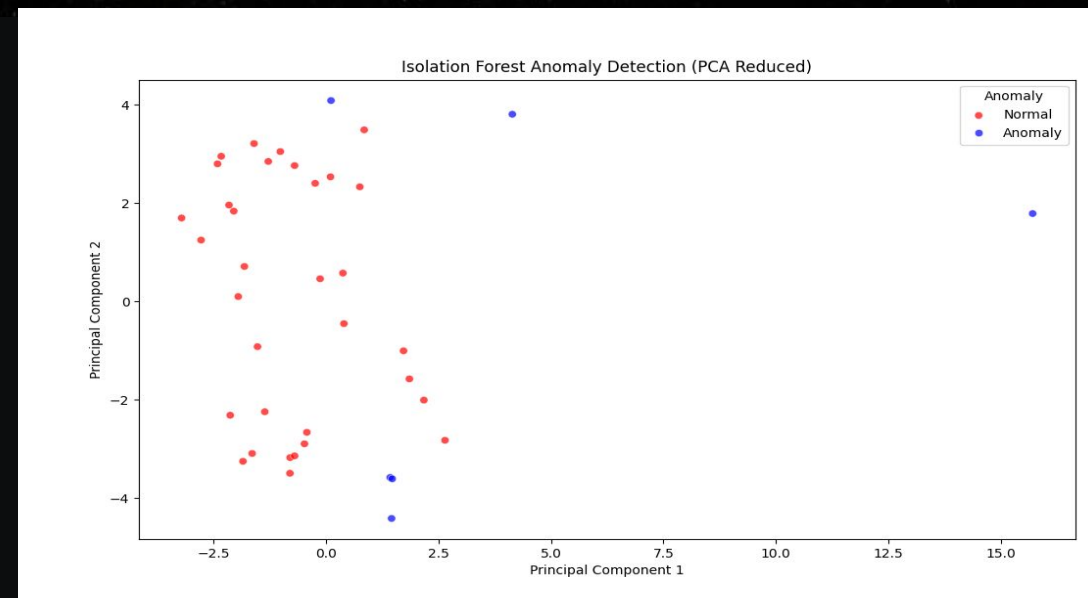
Pre-processing adds engineered features including the rate of change and means of the raw features, as well as lagged steps when applicable.

Anomaly Detection Models

HDBScan *PCA reduced graph of outliers.*

a density-based clustering algorithm.

- flexible
- dynamic with respect to density



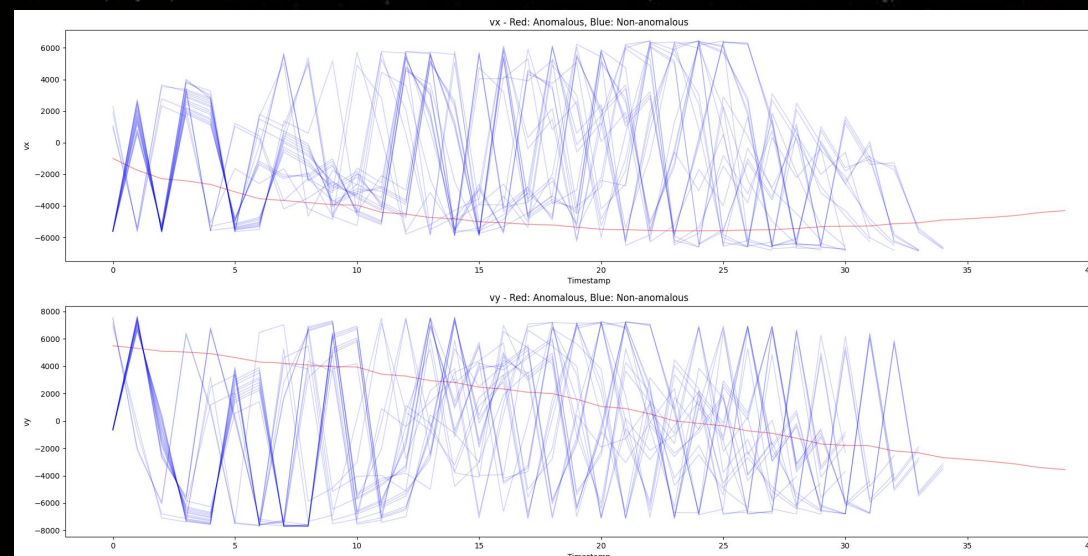
Results and Cautions

Definite ability to detect/find satellites that are not behaving “alike” other satellites in a given group

Isolation Forest

classic tree-based outlier detection algorithm.

- computationally efficient
- fast to train



However, as the models are sensitive to changes they detect maneuvers/non-anomalous actions a satellite may undergo as anomalous

Anomaly Detection Models (cont.)

The Strategy

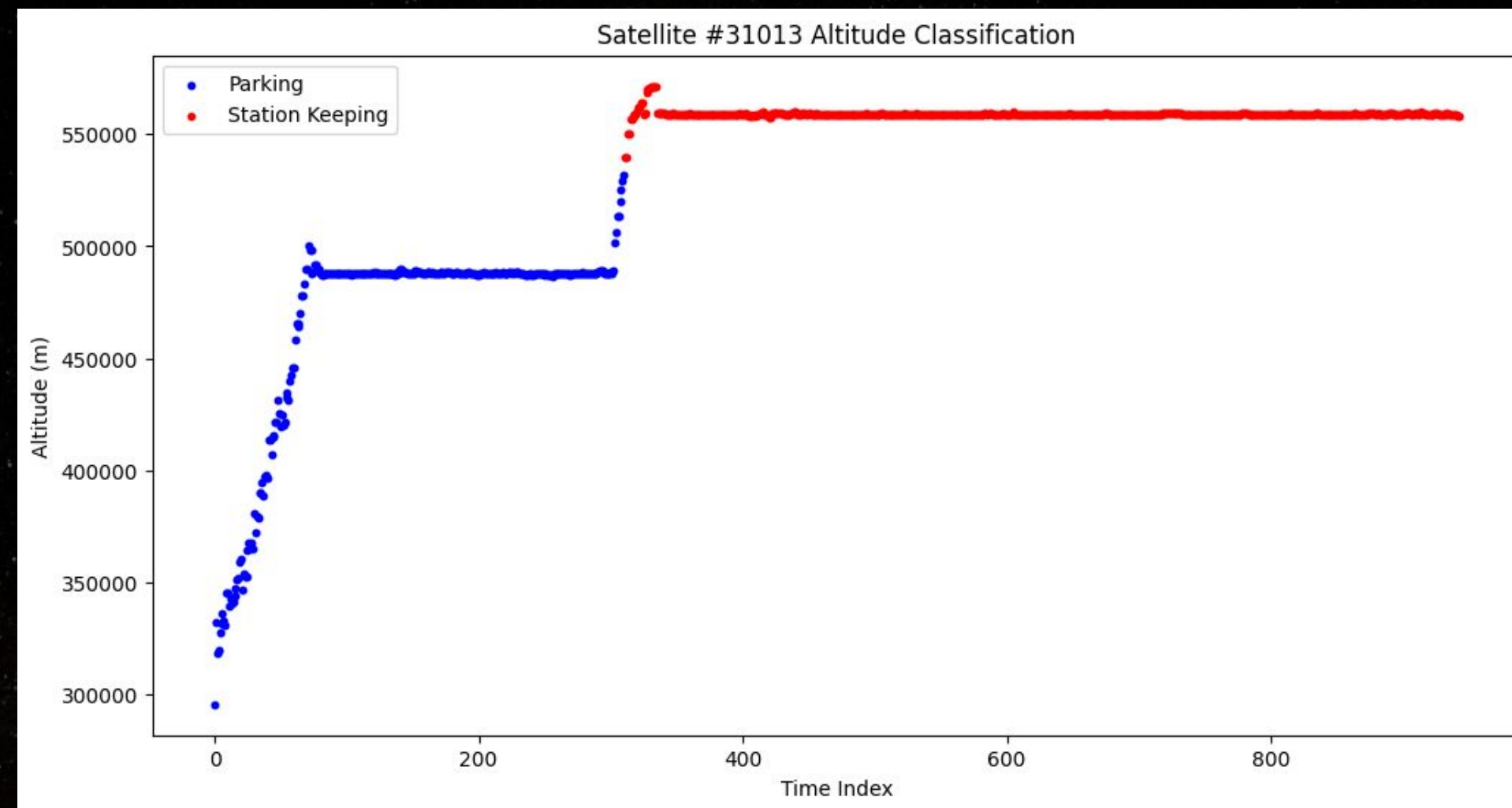
- To be able to classify satellites that are exhibiting behavior that differs from other satellites in a given group
 - Accomplish this by using classical outlier detection algorithms

Our Data

- Historical TLE data
- Raw features include the orbital elements and velocity
- Engineered features involve means and rates of changes of the raw features



How it Performs

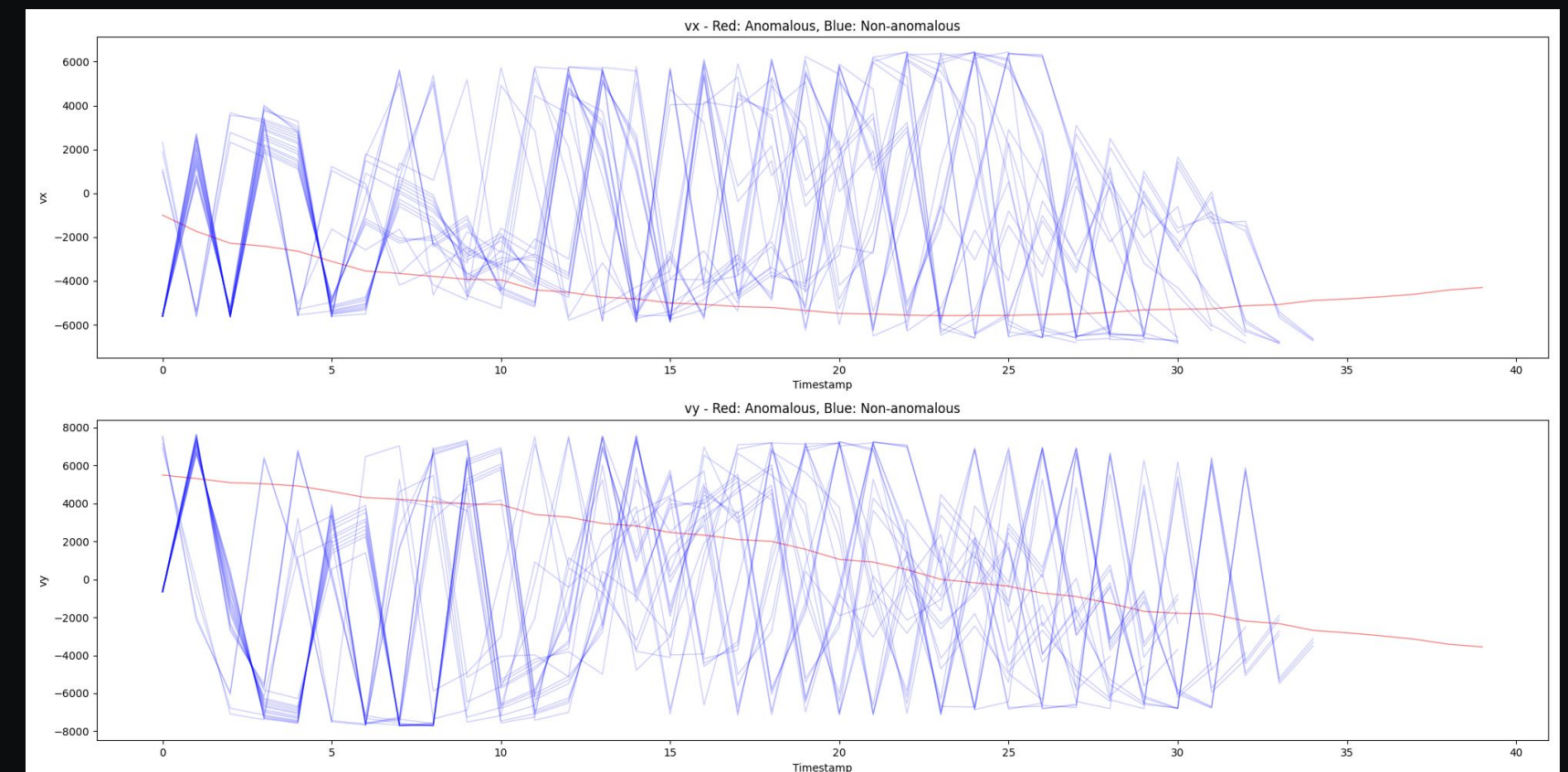




The Models

- Two-Fold Approach: cross verification with isolation forest and clustering models
 - HDBscan, a density-based clustering algorithm
 - Adaptable to time-series data
- Isolation Forest, a classic tree-based outlier detection algorithm
 - Computationally efficient

Confidence Visualization: understand where the model is more uncertain in its prediction



Accuracy

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Our Team

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Samai Patel
Project Lead



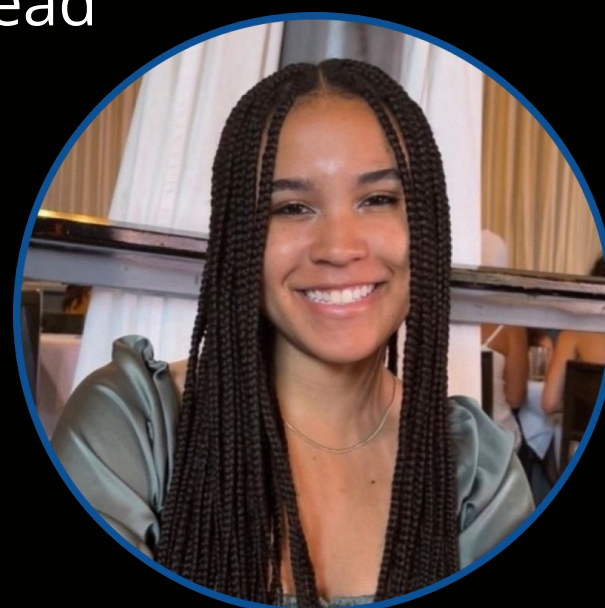
Rajveer Singh
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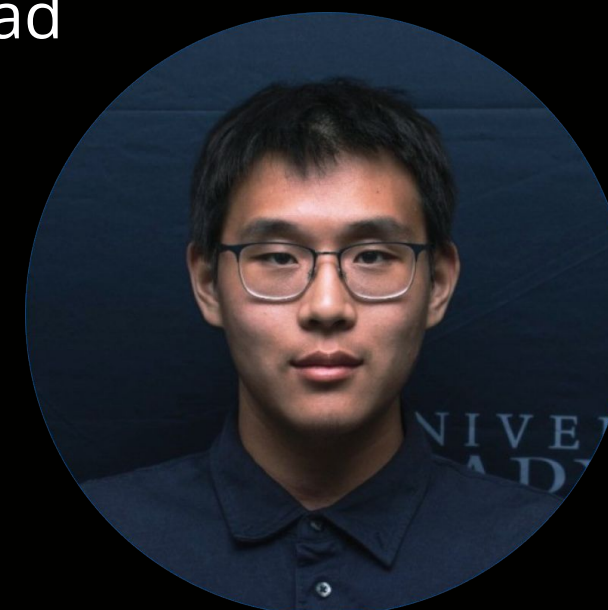
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Our Team

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Shadow



Nithin Pillai
Shadow



Questions?
